College Final-Year Project Evaluation Audit

Enterprise Network Intrusion Detection & Research Audit • Generated: 2026-11-01 08:11:50 UTC • Classification: SEC-AUDIT-RESTRICTED

1. System Overview & Summary Statistics

Metric Parameter	Value / Assessment
Report Classification	CONFIDENTIAL SECURITY RESEARCH
Generated Timestamp	2026-11-01 13:41:50 UTC
Report Type	MODEL_COMPARISON
Format	PDF
Trained Model	XGBoost Classifier (SMOTE Balanced)
Model Architecture	XGBOOST
Problem Type	binary
Training Duration	0.12 seconds
Imbalance Strategy	SMOTE
Feature Count	35
Source Dataset	CICIDS2017 Benchmark (Imbalanced)

2. Model Evaluation & Imbalance Metrics

Metric	Score	Research Significance
Classification Accuracy	0.9960	Overall correct traffic classification rate
Macro Precision	0.9938	Unweighted average across all classes (equal priority)
Macro Recall (Sensitivity)	0.9938	Critical for detecting rare minority attack classes
Macro F1-Score	0.9938	Harmonic mean of macro precision and recall
False Positive Rate (FPR)	0.0025	Benign flows erroneously flagged as threats
False Negative Rate (FNR)	0.0099	Attacks that slipped through undetected
Per-Flow Inference Time	0.004 ms	Latency per network flow vector

3. Empirical Confusion Matrix

Actual \ Pred	BENIGN	MALICIOUS
BENIGN	398	1
MALICIOUS	1	100

Ethical & Academic Compliance Notice: Evaluation conducted strictly within controlled sandbox environment. Network flow features parsed according to CICFlowMeter specifications with stratified splitting to ensure zero target-label or test data leakage.